

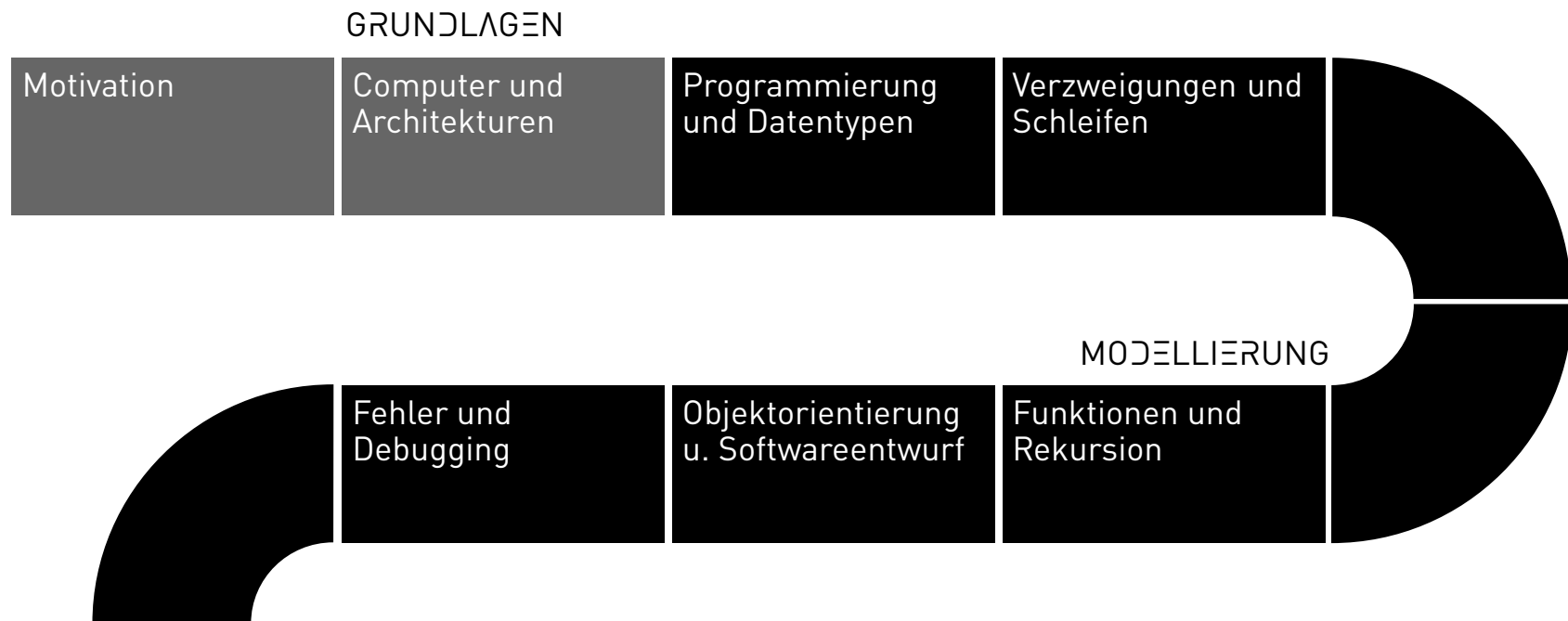
Programming Databases

Joern Ploennigs

Knowledge Pyramid

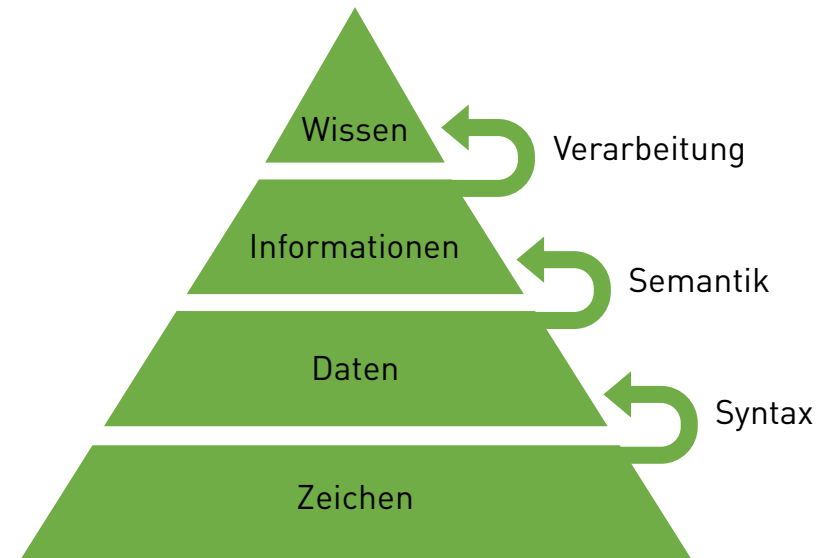
MIDJOURNEY: TOWER OF BABEL WITH MODERN CONSTRUCTION, REF. PIETER BRUEGEL

PROCESS



KNOWLEDGE PYRAMID

The Wissenspyramide is a model for representing the emergence of knowledge. The four element types: *Signs*, *Data*, *Information*, and *Knowledge* are represented pyramidically as four levels. Signs form the base and knowledge forms the apex of the pyramid.



CHARACTERS

Definition: Characters

A character encodes a single letter, a digit, a special character, or a control character.

- Example 1: German alphabet with umlauts and numbers, sentence separators and end-of-sentence marks Character set:

`abcdefghijklmnopqrstuvwxyz`

`ABCDEFGHIJKLMNOPQRSTUVWXYZ`

`äöüßÄÖÜ0123456789, : ; - . ! ?`

- Example 2: ASCII text (ISO 8859-1) is an encoding of 256 representable characters in 8 bits. (Originally there were only 128 characters in 7 bits.)



SYNTAX

Definition: Syntax

Formal rules that define how well-formed expressions are formed from a given character set.

- Example 1: German grammar - Sentence rules:
 - One or more words, each consisting of two or more alphanumeric characters
 - A space separates two words
 - **,** **:** **;** **-** separate sentence parts
 - **.** **!** **?** ends a sentence
- Example 2: HTML (web page):
 - Each line contains ASCII characters
 - Line breaks separate lines
 - A line contains text or an HTML tag that begins with **<** and ends with **>**



DATA



Definition: Data

A representation of information that can be interpreted repeatedly in a formalized manner, suitable for communication, interpretation, or processing.

- Example 1: Formally correct sentence. e.g., "I only understand Bahnhof."
- Example 2: HTML text file (*.html).



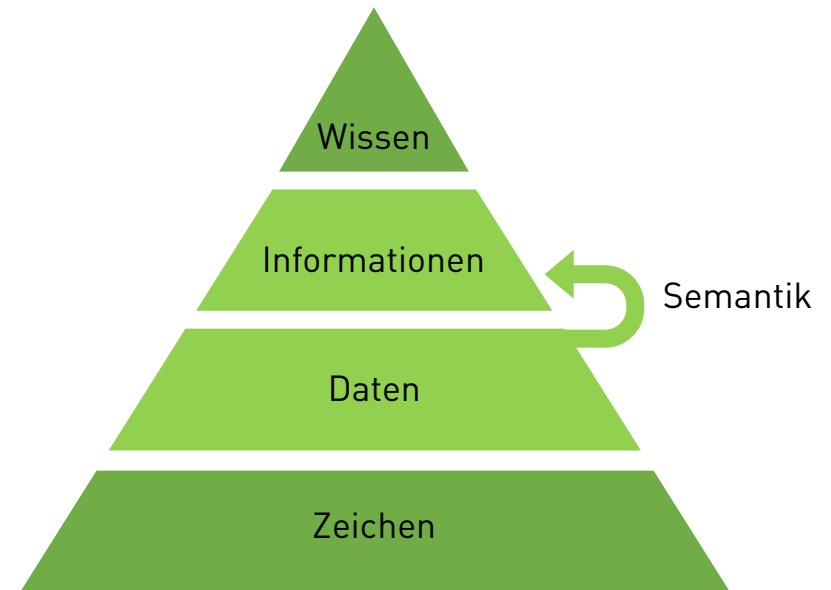
SEMANTICS



Definition: Semantics

Guidelines for interpreting data.

- Example 1: German dictionary. For example, the definition of "Bahnhof" in Duden:
 - The entire complex of a railway station with track facilities and associated buildings
 - a building belonging to the station with a [large] hall
- Example 2: HTML elements
 - `<p>` defines a paragraph
 - `<img>` defines an image
 - `<table>` defines a table
 - ...



INFORMATION

Definition: Information

Data that have meaning and are organized for a specific purpose. For example, information is a collection of data with accompanying explanations, interpretations, and other textual material that relate to a particular object, event, or process.

- Example 1: Correct German sentence e.g., "I have no idea what you're talking about."
- Example 2: HTML text file (*.html).



PROCESSING

Definition: Processing

Linking information through algorithms, experiences, values, and domain knowledge.

- Example 1: Germans know that the expression "Ich verstehe nur Bahnhof" is an idiom.
- Example 2: Web browser: A web browser interprets the HTML tags and renders a webpage from them



KNOWLEDGE

Definition: Knowledge

The collected, interconnected information about a specific subject, integrated with personal experience. Knowledge has a considerably higher level of complexity than information and is one of the company's most valuable assets. Through knowledge, decisions can be made and actions initiated.

- Example 1: "I don't understand a thing." This means not understanding anything or wanting to understand.
- Example 2: A fully rendered website



LECTURE HALL QUESTION

What are the signs shown in the image?



<https://www.planet-wissen.de/geschichte/antike/hieroglyphen>

LECTURE HALL QUESTION

What syntax is shown in the image?



<https://www.planet-wissen.de/geschichte/antike/hieroglyphen>

LECTURE HALL QUESTION

What is the semantic content of the image?



<https://www.planet-wissen.de/geschichte/antike/hieroglyphen>

LECTURE QUESTION

What processing is shown in the image?



<https://www.planet-wissen.de/geschichte/antike/hieroglyphen>

ANSWERS: HIEROGLYPHS

- Character: Each hieroglyph represents a character
- Syntax: The hieroglyphs are arranged in columns. We read them from top to bottom and from right to left.
- Semantics: Each hieroglyph has a context-specific meaning which can represent pictographs, phonetic signs, or determinatives.
- Processing: The hieroglyphs can only be correctly interpreted in their context.

a)	 „Mann“	 „Kind“	 „Rind“	 „Haus“
b)	 „trinken“	 „schlagen“	 „fliegen“	 „weinen“
c)	 „mähen“	 „schreiben“	 „hacken“	 „schneiden“
d)	 „Auge“ und „sehen“	 „Mann“ und „sitzen“	 „Schiff“ und „fahren“	 „Last“ und „tragen“
e)	 „Tag“ (Sonne)	 „Monat“ (Mond)	 „Stunde“ (Stern)	 „Arbeit“ (Biene)

SYNTAX OF A PROGRAMMING LANGUAGE

Definition: Syntax in Programming Languages

A formal set of rules that defines how the computer should interpret the programming language. The syntax typically includes rules for defining commands (keywords), variables (values), and code blocks (statements).

Syntax Beispiel	Bedeutung
1234	Zahlen Wert
True / False	Booleschen Werte
"Text"	Zeichenkette
# Ignoriere	Kommentar
+ - / *	Operatoren
if True: print("Hello") break	Einrückung eines Code-Blocks in Python. Achtung! Alle Zeilen eines Code-Blocks müssen gleich eingerückt sein. Eine "if" Bedingung wird immer mit ":" abgeschlossen

SEMANTICS OF A PROGRAMMING LANGUAGE



Definition: Semantics in programming languages

Definition of the meaning of language elements and the permissible combinations of elements.

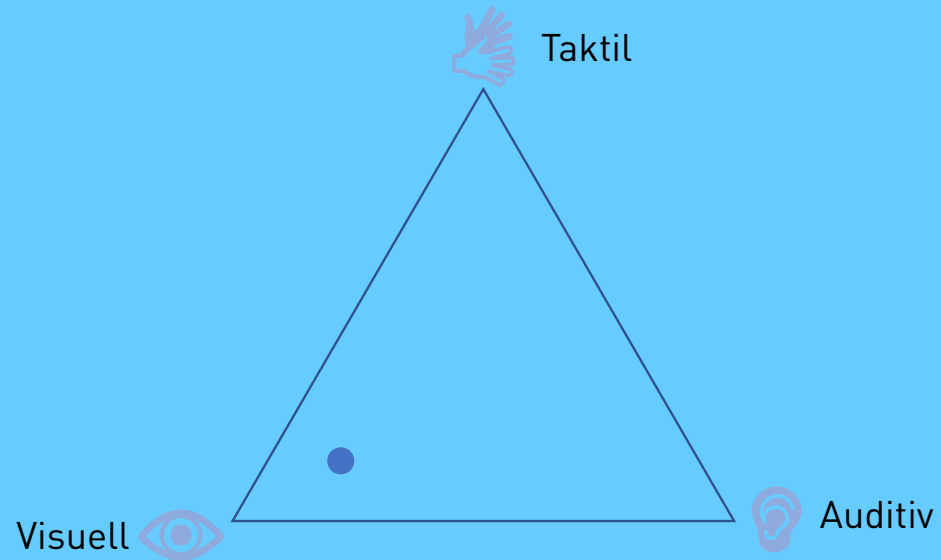
Semantik Beispiel	Bedeutung
<code>print("Hello")</code>	Gibt die Zeichenkette "Hello" aus
<code>exit()</code>	Beendet ein Programm
<code>pass</code>	Tut gar nix
<code>break</code>	Bricht einen Code-Block ab
<code>while</code>	Wiederhole den folgenden Code-Block

LESSON LEARNED

©Midjourney: Six Sense

What senses do we have?

LESSONS LEARNED - LEARNING STYLES



- Each of us has preferred sensory channels
- We tend to receive and transmit information through these channels
- This can be used intentionally for learning
- The most effective learning occurs when you use multiple channels

LESSONS LEARNED – LEARNING STYLE TESTING

- Using the following test, you can determine which learning style you tend to have. For each question, without overthinking it, place a cross in one of the numbered boxes.
- Now tally how often you marked answers from categories 1, 2, and 3. Enter the totals for each category in the table below.

Wie lernen Sie am liebsten eine neue Sprache?	
Ich suche mir gute Lehrbücher und lese Romane in der Fremdsprache.	1
Ich sehe mir am liebsten fremdsprachige Filme an und höre Hörbücher.	2
Ich suche mir fremdsprachige Bekannte und reise so oft wie möglich in das Land.	3
Welche Prüfungsform bevorzugen Sie?	
Ich bevorzuge mündliche Prüfungen.	2
Ich bevorzuge schriftliche Prüfungen.	1
Ich bevorzuge praktische Prüfungen.	3
Wie vermitteln Sie Wissen am liebsten?	
Ich zeige jemandem, wie ein Gerät funktioniert oder was er tun sollte.	3
Ich schreibe mein Wissen am liebsten auf.	1
Ich halte gerne Referate/Vorträge.	2
Wie lernen Sie die Funktionen eines neuen Handys kennen?	
Ich bitte jemanden, mir die Funktionen zu erklären.	2
Ich lese die Bedienungsanleitung.	1
Ich probiere alle Funktionen direkt aus.	3
Was macht Ihnen als Zuhörer bei Vorträgen/Workshops am meisten Spaß?	
Die praktischen Übungen.	3
Ich höre zu und stelle viele Fragen.	2
Ich schreibe aufmerksam mit und lese mir die Mitschrift hinterher durch.	1
Wie bereiten Sie sich auf Prüfungen am liebsten vor?	
Ich besorge mir Lehrbücher mit Übungen und Tests.	3
Ich erstelle eigene Dokumente, indem ich die wichtigsten Informationen aufschreibe.	1
Ich bitte jemanden, mich abzufragen.	2
Was halten Sie von Hörbüchern?	
Hörbücher sind für mich eine optimale Lernhilfe.	2
Ich ziehe herkömmliche Bücher vor.	1
Ich lerne lieber durch Ausprobieren.	3

LESSONS LEARNED — TAILORING TO LEARNING STYLES

The tactile learner learns best when they can touch or try things out.

- Try things out
- Touch things
- Move while learning
- Take a walk
- Create flashcards
- Make models
- Create study posters
- Write by hand

The auditory learner learns best when they hear something.

- Phrase the material as short learning statements
- Read them aloud to yourself
- Record the material on your smartphone and replay it via headphones
- Talk with others
- Find study groups
- Find quiet places without distractions

The visual learner learns best when they can see something.

- Take notes in bullet form
- Create illustrations
- Look for suitable mental images
- Use visual aids
- Visualize connections in sketches, pictures, and summaries
- Review overviews and diagrams repeatedly

Questions?

